

TIRANA RINAS, Albania

WMO: 136150

Lat: 41.415N

Lon: 19.721E

Elev: 38

StdP: 100.86

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.3	-1.8	-11.8	1.4	3.8	-8.5	1.8	3.8	9.8	11.2	8.3	10.6	1.2	170	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.1	35.1	22.5	33.8	22.7	32.7	22.9	27.3	29.9	26.1	29.2	25.0	28.9	3.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	28.8	25.2	20.4	27.6	24.0	19.0	26.5	86.8	30.6	81.5	28.4	76.8	28.0	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.4	5.5	-6.2	37.7	2.0	1.7	-7.7	38.9	-8.9	39.9	-10.0	40.8	-11.5	42.1		
			-6.8	28.1	2.2	2.9	-8.4	30.2	-9.7	31.9	-11.0	33.5	-12.6	35.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	7.1	8.8	11.5	14.9	18.5	22.8	25.2	21.6	17.1	12.7	8.2
		DBStd	6.98	3.37	3.27	2.46	2.38	2.68	2.92	2.00	1.92	2.79	3.29	3.44
		HDD10.0	259	99	55	14	1	0	0	0	0	1	13	78
		HDD18.3	1513	349	267	213	105	31	2	0	6	57	169	314
		CDD10.0	2528	9	22	59	149	264	385	472	347	220	94	21
		CDD18.3	739	0	0	3	37	137	214	226	103	18	1	0
		CDH23.3	7443	0	1	2	46	324	1341	2292	2390	903	141	4
		CDH26.7	3281	0	0	0	6	68	546	1103	1219	324	13	0
(14)	Wind	WSAvg	2.1	2.1	2.5	2.5	2.1	2.0	2.0	2.0	1.9	1.7	1.9	2.0
(15)	Precipitation	PrecAvg	911	75	82	92	92	49	46	30	32	82	95	126
(16)		PrecMax	1297	203	205	183	168	114	127	89	154	279	228	290
(17)		PrecMin	535	11	0	2	14	18	0	0	0	1	48	19
(18)		PrecStd	200	50	53	56	43	29	38	24	39	65	58	62
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	17.6	20.8	22.8	27.5	30.9	35.2	36.8	37.1	34.0	28.2	23.8	19.2
(20)		MCWB	12.1	12.3	12.7	17.8	22.3	23.2	22.3	22.1	21.2	18.6	17.8	15.7
(21)		DB	15.9	18.1	20.8	24.3	28.7	33.2	34.9	35.2	32.1	26.4	21.8	17.1
(22)		MCWB	11.6	12.4	12.9	16.8	21.9	24.0	23.0	22.1	21.1	18.8	16.2	13.6
(23)		DB	14.3	16.2	19.1	22.8	26.9	31.9	33.6	34.1	30.2	24.9	20.2	15.8
(24)		MCWB	11.1	11.5	13.0	15.8	20.4	23.6	23.1	22.3	20.9	18.7	15.1	12.2
(25)		DB	13.0	14.9	17.8	21.1	25.0	30.1	32.2	32.8	28.4	23.2	18.8	14.2
(26)		MCWB	10.0	10.6	12.2	15.4	19.0	22.8	23.1	22.8	20.5	18.2	14.8	11.1
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	15.4	16.1	17.7	20.8	25.6	29.7	29.7	28.6	26.1	23.9	20.0	17.2
(28)		MCDB	15.9	16.9	18.4	22.2	27.9	31.7	31.0	30.9	27.9	24.9	21.1	18.2
(29)		WB	13.4	14.0	15.6	19.3	23.6	27.0	27.4	26.7	24.2	22.0	17.9	15.1
(30)		MCDB	14.2	16.0	17.0	21.5	26.2	30.0	30.0	29.8	27.4	23.2	19.6	15.9
(31)		WB	11.8	12.5	14.2	18.0	22.1	25.3	26.1	25.6	22.9	20.1	16.5	13.4
(32)		MCDB	13.2	15.3	17.0	20.8	24.9	28.6	29.3	29.5	26.6	22.4	18.6	14.8
(33)		WB	10.6	11.3	13.1	16.8	20.4	23.8	24.9	24.4	21.7	18.8	15.4	12.0
(34)		MCDB	12.4	14.1	16.3	19.8	23.5	27.7	28.9	29.0	25.9	22.0	17.7	13.2
(35)	Mean Daily Temperature Range	MDBR	10.5	10.5	11.3	11.5	11.5	11.9	13.3	14.1	12.9	12.3	11.0	11.1
(36)		5% DB	11.9	12.4	14.1	14.4	13.9	14.3	15.2	16.1	15.0	14.2	12.2	11.9
(37)		MCWBR	8.2	7.6	8.0	8.2	7.8	7.1	6.5	6.2	6.5	7.8	7.5	8.2
(38)		5% WB	9.8	10.3	11.4	12.4	12.7	12.6	12.9	13.3	12.9	11.7	10.5	9.4
(39)	Clear-Sky Solar Irradiance	MCWBR	8.1	7.5	8.1	9.2	9.3	8.7	8.2	7.8	7.8	8.7	8.2	7.5
(40)		taub	0.346	0.373	0.421	0.475	0.463	0.439	0.441	0.460	0.439	0.406	0.374	0.348
(41)		taud	2.410	2.313	2.178	2.047	2.149	2.272	2.259	2.199	2.246	2.314	2.357	2.404
(42)		Ebn at Noon	794	824	819	799	820	840	834	804	793	776	749	756
(43)	All-Sky Solar Radiation	Edh at Noon	82	104	133	162	150	133	133	138	123	103	86	76
(44)		RadAvg	1.78	2.58	3.81	4.96	6.20	7.13	7.36	6.45	4.69	3.24	2.05	1.58
(45)		RadStd	0.19	0.38	0.50	0.50	0.44	0.41	0.30	0.33	0.34	0.30	0.19	0.26

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)		+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-80	N/A	N/A

Nomenclature: See separate page